Statistics of Financial Markets

Stable Distributions

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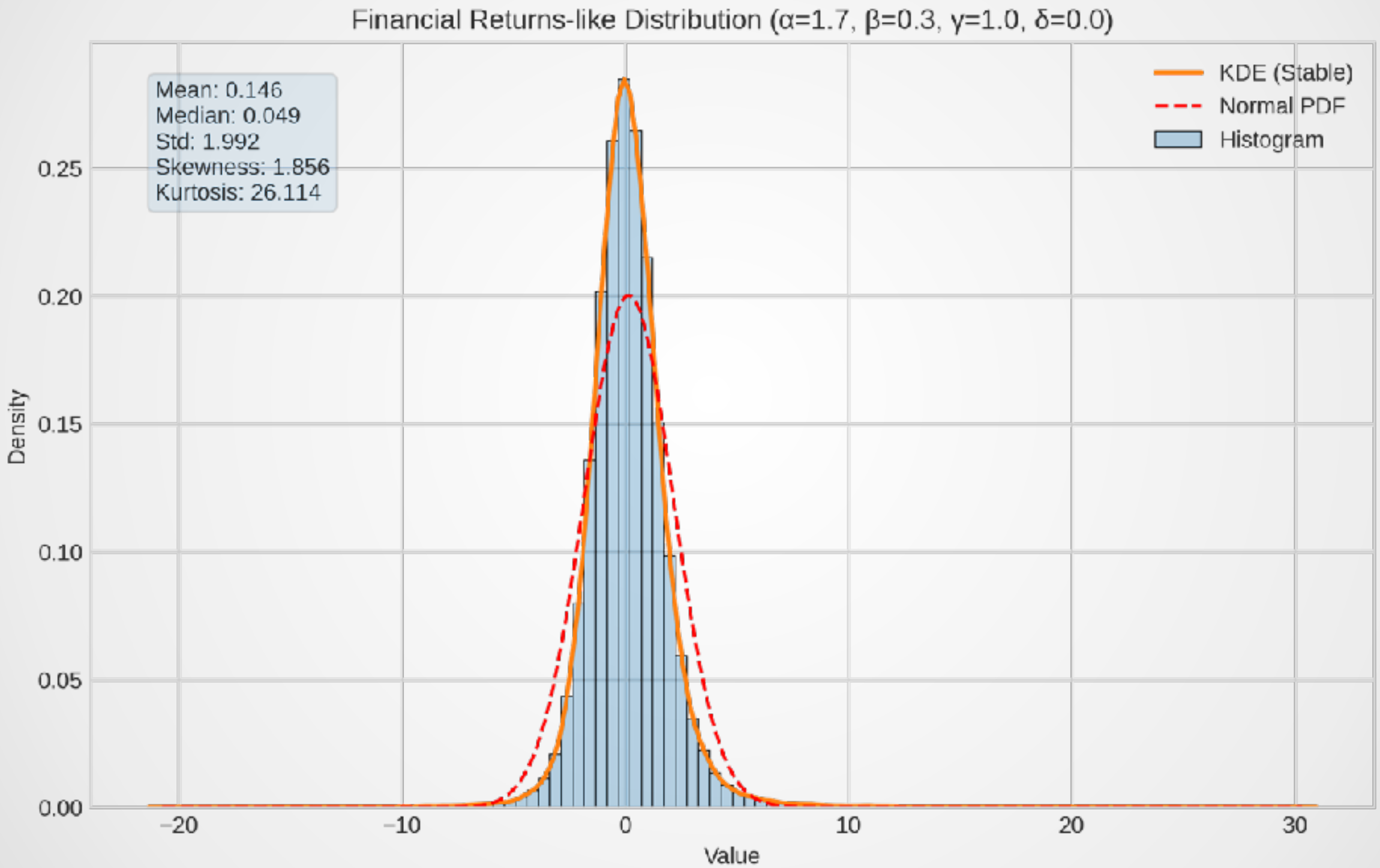
Why Study Stable Distributions?

- Real-World Data ≠ Gaussian
  - Financial returns, insurance claims, environmental data, and telecommunication noise often display heavy tails and asymmetry.
  - Real-world data frequently violates the assumptions of finite variance and Gaussianity.
- Theoretical motivation
  - Stable distributions emerge as limit laws under the Generalized Central Limit Theorem (GCLT).
  - They preserve their distributional form under addition and scaling
- Practical importance
  - Risk modeling, extreme value analysis (EVT), and robust signal processing.
  - Offer a unifying framework that extends beyond the Gaussian model.

Why Study Stable Distributions?

- Rich Mathematical Structure
  - Elegant formulation via characteristic functions
  - Deep links with Fourier analysis, stochastic processes, and fractal geometry
- Applications Across Disciplines
  - Finance: heavy-tailed asset returns, VaR, stress testing
  - Physics: Lévy flights, anomalous diffusion
  - Signal Processing: impulsive noise, clutter modeling

Stable Distributions: Examples



Outline

1. Motivation ✓
2. Introduction
3. Definition
4. Properties
5. Examples
6. Parameterization
7. Estimation
8. Simulation
9. Conclusions and Homework
10. References

Stable Distributions

- A class of probability distributions closed under convolution (i.e., summation of i.i.d. variables).
- Includes Gaussian distribution as a special case ($\alpha = 2$).
- Allows for skewness ($\beta \neq 0$) and heavy tails ($\alpha < 2$).
- Defined via their characteristic function since most do not have closed-form PDFs.
- Applications in modeling
  - Rare/extreme events
  - Self-similar and fractal phenomena
  - Systems lacking finite variance or higher-order moments

Historical Development

Year	Contributor	Contribution
1925	Paul Lévy	Introduced stability and defined the class
1938	A. Khintchine	Explored limit properties and representations
1954	Gnedenko & Kolmogorov	Formalized Generalized CLT and domains of attraction
1960s	B. Mandelbrot	Applied to finance and fractal modeling
1971	William Feller	Integrated stable laws into modern probability theory
1997–	John Nolan	Developed numerical algorithms and software tools

What Are Stable Distributions?

- A generalization of the normal distribution family
- Allow for
  - Heavy tails (infinite variance)
  - Skewness (asymmetry)
- Defined by stability under addition:
  - $aX_1 + bX_2 \stackrel{d}{=} cX + d$
- Used when Gaussian models underestimate tail risk

Defining Stability

- A random variable X is stable if:
  - $aX_1 + bX_2 \stackrel{d}{=} cX + d$ for any $a, b > 0$, where $X_1, X_2 \stackrel{iid}{\sim} X$.
  - $c = \left(a^\alpha + b^\alpha\right)^{1/\alpha}$
- For α-stable distributions
  - This property defines the closure under linear combinations, fundamental to the concept of distributional stability.
  - Particularly important in modeling sums of many small, independent shocks or innovations.

Generalized Central Limit Theorem (GCLT)

- Let $X_i, i = 1..n$ be i.i.d. random variables with a common distribution that has heavy tails (i.e., infinite variance or slowly decaying tails).
- Then, there exist sequences of constants $c_n > 0$ and $d_n \in \mathbb{R}$ such that:
  - $$\frac{\sum_{i=1}^{n} X_i - d_n}{c_n} \xrightarrow{d} X \sim S(\alpha, \beta, \gamma, \delta).$$
- Classical CLT
  - Converges to Gaussian when variance is finite ($\alpha = 2$).
- GCLT
  - Converges to an α-stable law when variance is infinite ($\alpha < 2$).
- Implication
  - Stable distributions generalize the central limit behavior of sums
  - Foundation for modeling sums of extreme or heavy-tailed observations
  - The foundation of stable models in risk and heavy-tailed phenomena.

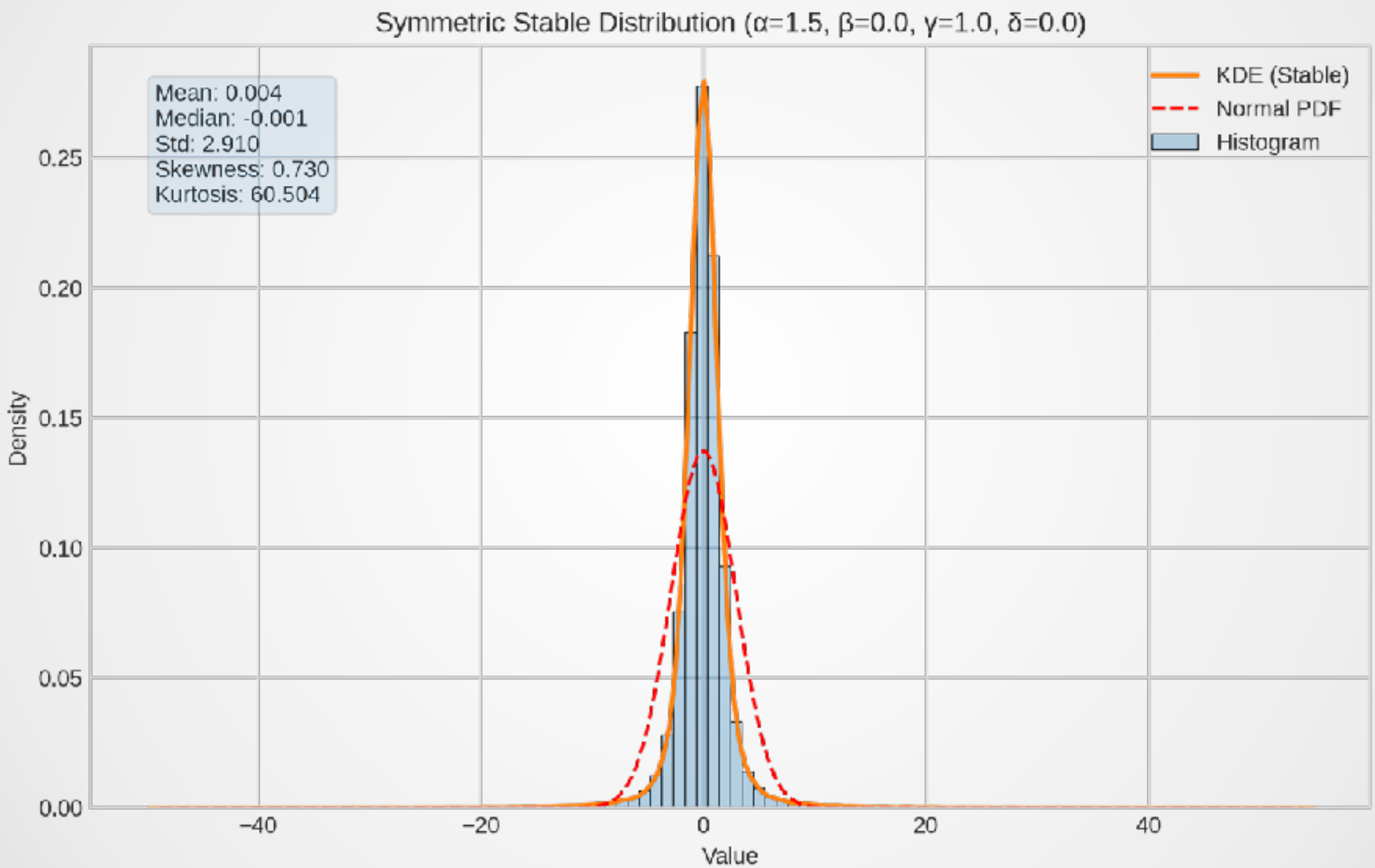
Parameterization Overview

- A stable distribution is denoted: $S(\alpha, \beta, \gamma, \delta)$
  - α, $(0 < \alpha \leq 2)$: stability index or tail exponent
  - β, $(-1 \leq \beta \leq 1))$: skewness parameter
  - γ, $(\gamma > 0)$: scale parameter
  - δ, $(\delta \in \mathbb{R})$: location parameter
- Interpretation
  - $\alpha = 2$: Normal distribution
  - $\beta = 0$: Symmetric distribution
  - Smaller α implies heavier tails and lower moment existence

Effects of parameters

- α
  - Controls tail thickness
  - As $\alpha \to 0$, tails become extremely heavy
- β
  - Adjusts asymmetry of the distribution
  - $\beta = -1$: heavy left tail
  - $\beta = 1$: heavy right tail
- γ
  - Analogous to standard deviation
  - Scales the distribution
- δ
  - Controls central location
  - Behaves like the mean when$\alpha > 1$

Stable Distributions: Examples



Characteristic Function (Parameterization A)

- Let $X \sim S_A(\alpha, \beta, \gamma, \delta)$
- The characteristic function (CF) $\phi(t) = \mathbb{E}[e^{itX}]$
  - If $\alpha \neq 1$: $\phi(t) = \exp\left\{-\gamma^\alpha |t|^\alpha \left[1 - i\beta \cdot \text{sign}(t) \cdot \tan\left(\frac{\pi\alpha}{2}\right)\right] + i\delta t\right\}$
  - If $\alpha = 1$: $\phi(t) = \exp\left\{-\gamma |t| \left[1 + i\beta \cdot \frac{2}{\pi} \cdot \text{sign}(t) \cdot \ln|t|\right] + i\delta t\right\}$
- The characteristic function uniquely defines a probability distribution.
- For stable distributions, the PDF is often unknown or intractable, so the CF is used instead.
- Parameters control behavior
- Useful for simulation (inverse transform methods), fitting, and deriving properties.

Characteristic Function (Parameterization B)

- Let $X \sim S_B(\alpha, \beta, c, \mu)$
- The characteristic function $\phi(t) = \mathbb{E}[e^{itX}]$
  - If $\alpha \neq 1$:

$$\phi(t) = \exp\left\{ -|ct|^{\alpha} \left[1 - i\beta \cdot \mathrm{sign}(t) \cdot \tan\left(\frac{\pi\alpha}{2}\right) \right] + i\mu t \right\}$$

  - If $\alpha = 1$: $\phi(t) = \exp\left\{ -|ct| \left[1 + i\beta \cdot \frac{2}{\pi} \cdot \mathrm{sign}(t) \cdot \ln|ct| \right] + i\mu t \right\}$
- Parameters c, μ are alternative scale and location parameters.
- Frequently used in mathematical proofs and simulations based on Zolotarev's representation.

Mapping Between Parameterizations

- Stable distributions may be expressed in two parameterizations
  - Standard (A): $S_A(\alpha, \beta, \gamma, \delta)$
  - Alternative (B): $S_B(\alpha, \beta, c, \mu)$
- Mapping from A to B
  - If $\alpha \neq 1 \succ c = \gamma, \mu = \delta$
  - If $\alpha = 1 \succ c = \gamma, \mu = \delta - \frac{2}{\pi}\beta\gamma \ln(\gamma)$
- Mapping from B to A
  - If $\alpha \neq 1 \succ \gamma = c, \delta = \mu$
  - If $\alpha = 1 \succ \gamma = c, \delta = \mu + \frac{2}{\pi}\beta c \ln(c)$
- **Note: in the following we use the standard parameterization**

  $S_A(\alpha, \beta, \gamma, \delta)$

Probability Density Function (PDF)

- In general, stable distributions do not have closed-form PDFs except for specific cases (Gaussian, Cauchy, Lévy).
- The PDF can be computed via the inverse Fourier transform:

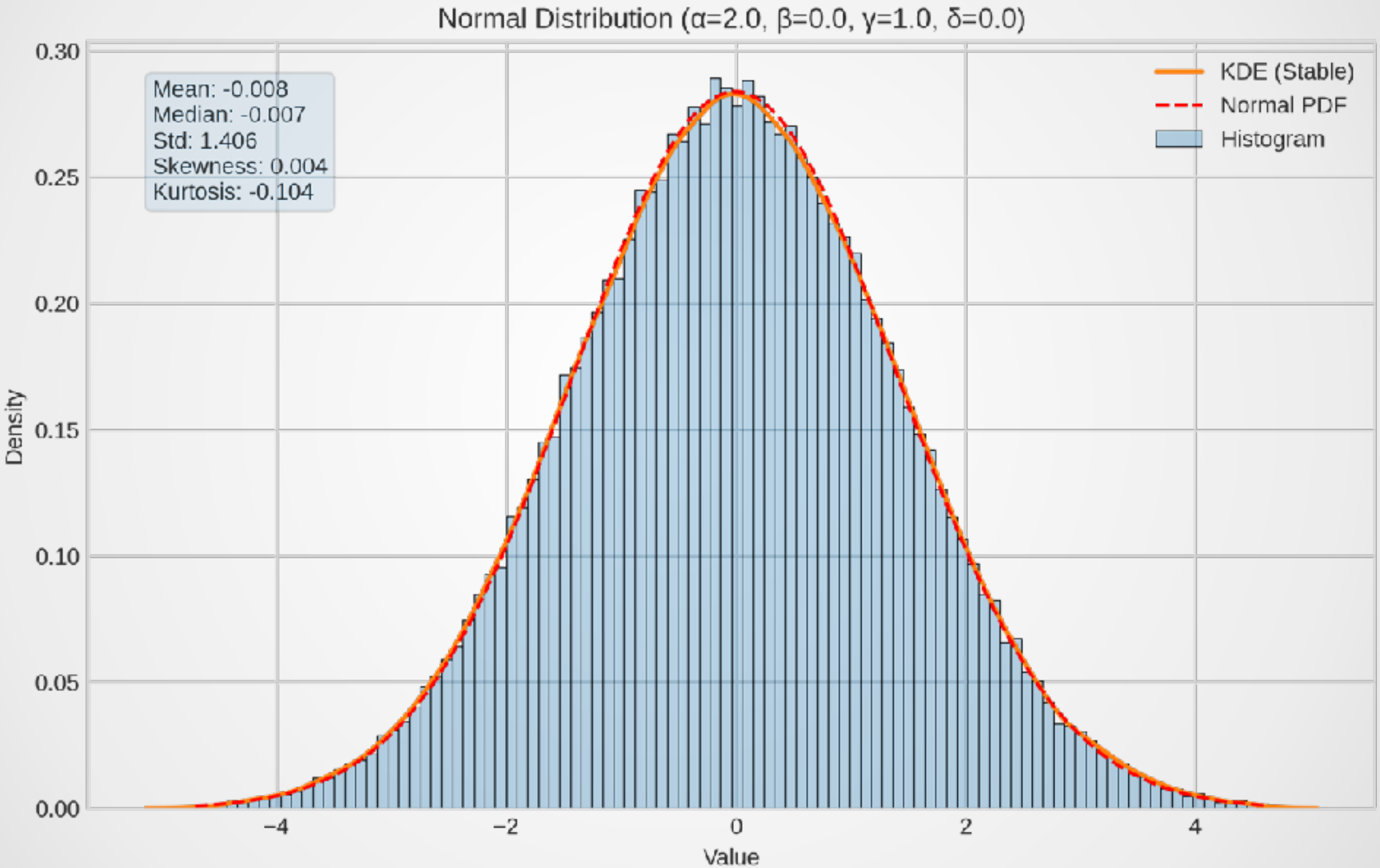
$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itx} \phi(t) dt$$

- Numerical techniques
  - Inversion of the characteristic function
  - Fast Fourier Transform (FFT)
  - Series expansions (e.g., Zolotarev's formula)
  - Computational tools: pystable, levy_stable, libstable4u, John Nolan's software

Special Case – Gaussian Distribution

- When $\alpha = 2$, the stable distribution becomes normal: $X \sim N\left(\delta, 2\gamma^2\right)$.
- Characteristic function: $\phi(t) = \exp\left(-\gamma^2 t^2 + i\delta t\right)$.
- PDF: $f(x) = \dfrac{1}{\sqrt{4\pi\gamma^2}} \exp\left(-\dfrac{(x-\delta)^2}{4\gamma^2}\right)$
- Properties
  - Symmetric, light-tailed
  - All moments exist
  - The only stable distribution with finite variance.

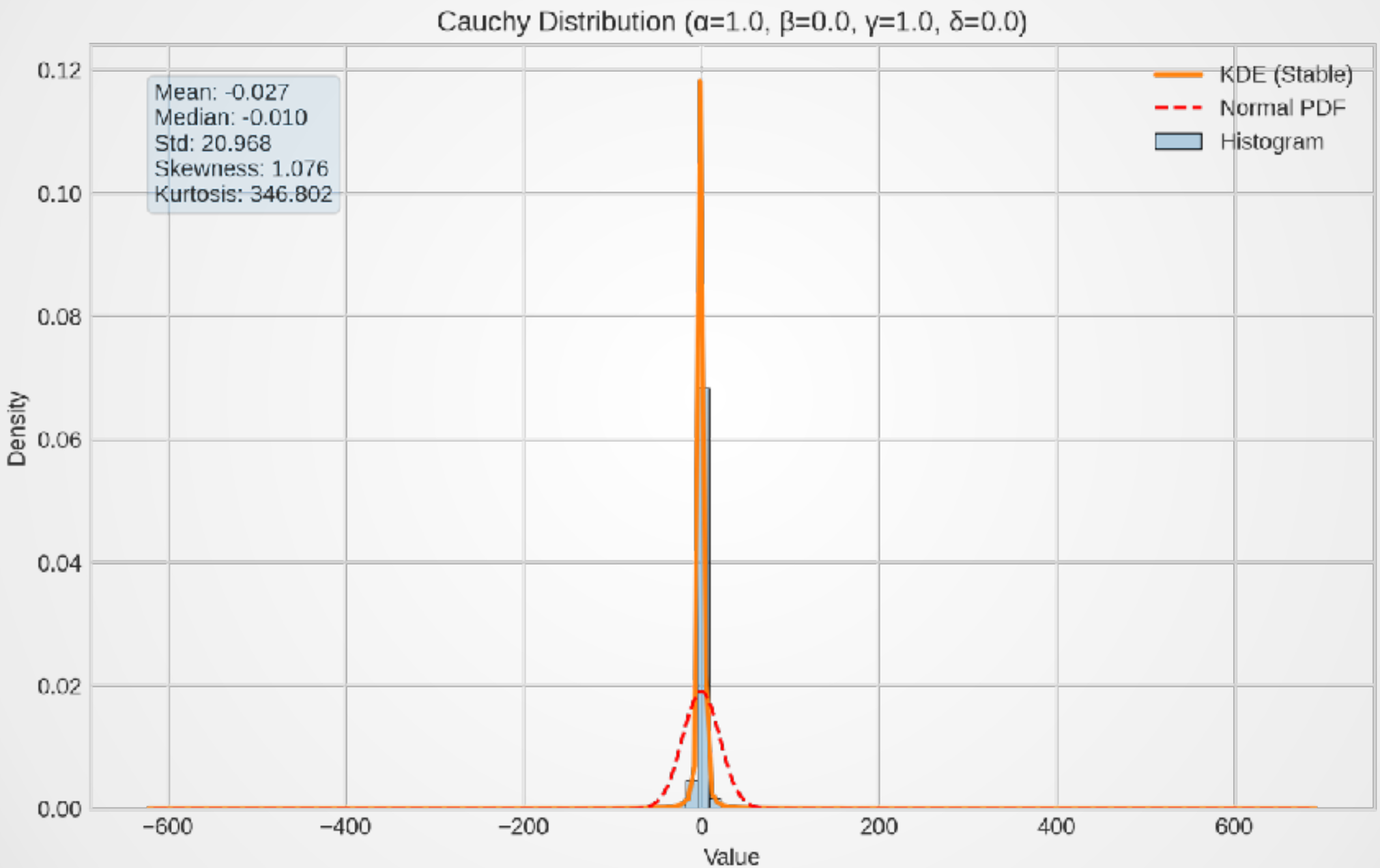
Stable Distributions: Gaussian



Special Case – Cauchy Distribution

- When $\alpha = 1,\ \beta = 0$, we get the Cauchy distribution: $X \sim \mathrm{Cauchy}(\delta, \gamma)$
- PDF: $f(x) = \dfrac{1}{\pi\gamma\left[1 + \left(\frac{x-\delta}{\gamma}\right)^2\right]}.$
- Properties
  - Symmetric, heavy-tailed
  - No mean, variance, or higher moments
  - Ratio of two standard normals is Cauchy-distributed

Stable Distributions: Cauchy



Special Case – Lévy Distribution

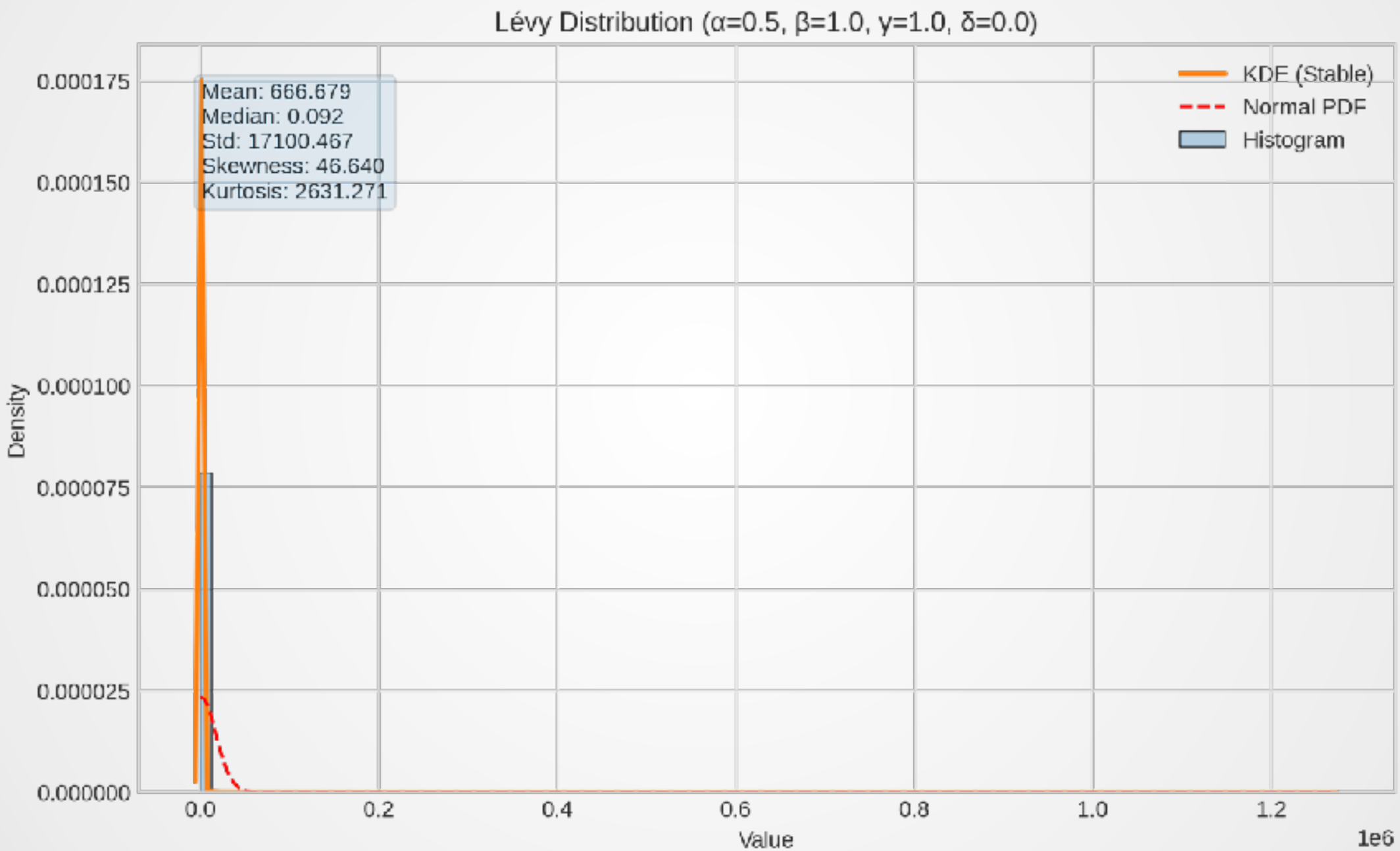
- When $\alpha = 0.5,\ \beta = 1$, we get the Lévy distribution:

  $$X \sim S(0.5,1,\gamma,\delta), \quad X > \delta.$$

- PDF: $f(x) = \sqrt{\dfrac{\gamma}{2\pi}} \cdot \dfrac{e^{-\gamma/(2(x-\delta))}}{(x-\delta)^{3/2}}, \quad x > \delta$

- Applications
  - First passage times of Brownian motion
  - Extreme value theory

Stable Distributions: Examples



Tail Behavior of Stable Distributions

- For $0 < \alpha < 2$, stable distributions exhibit power-law tails:
  - $P(X > x) \sim C_\alpha(1+\beta)x^{-\alpha}, \quad P(X < -x) \sim C_\alpha(1-\beta)x^{-\alpha}$
  - $C_\alpha = \dfrac{1-\alpha}{\Gamma(2-\alpha)\cos(\pi\alpha/2)}$
- The smaller α, the heavier the tails
- β skews the distribution: more mass on left or right tail
- Tail decay is polynomial, not exponential (unlike Gaussian)
- High impact, low probability events are more likely

Moments and Their Existence

- Existence of moments depends on the value of α:
  - $E[|X|^p] < \infty$ for $0 < p < \alpha$
  - $E[|X|^p] = \infty$ for $p \geq \alpha$
  - Mean exists only if $\alpha > 1$
  - Variance exists only if $\alpha = 2$(Gaussian case)
  - For Cauchy ($\alpha = 1$): no mean, no variance
  - For Lévy ($\alpha = 0.5$): even $E[X]$does not exist
- This property makes stable distributions suitable for extreme modeling, but poses challenges for inference

Parameter Estimation Techniques

- No closed-form PDF for general stable distributions
- Maximum Likelihood Estimation (MLE)
  - Uses numerical inversion of the characteristic function via Fast Fourier Transform (FFT)
  - Log-likelihood $\ell(\theta) = \sum_{i=1}^{n} \log f\left(x_i; \theta\right)$, where $f(x)$ is computed via FFT inversion
  - Requires good initial estimates and regularization techniques
- Quantile Matching Estimation (QME)
  - Minimizes squared differences between empirical and theoretical quantiles:

$$\min_{\theta} \sum_{k=1}^{K} \left(Q_k^{\text{empirical}} - Q_k^{\text{theoretical}}(\theta) \right)^2$$

  - Robust for small samples and heavy tails

Parameter Estimation Techniques

- Characteristic Function (CF) Based Estimation
  - Define an objective $J(\theta) = \int |\hat{\phi}(t) - \phi(t;\theta)|^2 w(t)dt$
  - Minimized using numerical integration and nonlinear optimization
  - Advantage: does not require evaluating PDF
- McCulloch's Quantile Estimation Method
  - Uses five selected quantiles to estimate α, β, γ, δ.
  - Based on pre-tabulated values from simulations
  - Very fast and easy to implement, good for large datasets

Parameter Estimation Techniques

- Koutrouvelis Regression Method
  - Based on the empirical characteristic function (ECF)

$$\hat{\phi}(t) = \frac{1}{n}\sum_{j=1}^{n} e^{itX_j}$$

  - Stage 1: Linear regression for α and γ:

$$\log(-\log|\hat{\phi}(t)|) = \alpha \log|t| + \alpha \log\gamma + \epsilon_1(t).$$

  - Stage 2: Regression for β and δ via phase:

$$\frac{\arg(\hat{\phi}(t))}{t} = -\beta\gamma^{\alpha}\tan\left(\frac{\pi\alpha}{2}\right)|t|^{\alpha-1} + \delta + \epsilon_2(t).$$

  - Fast, widely used, effective for moderate to large samples

Chambers-Mallows-Stuck (CMS) Method

- Simulates $X \sim S(\alpha, \beta, \gamma, \delta)$ using Parameterization A:

$$X = \gamma \cdot \frac{\sin(\alpha(U+\zeta))}{\cos(U)^{1/\alpha}} \left[\frac{\cos(U - \alpha(U+\zeta))}{W} \right]^{(1-\alpha)/\alpha} + \delta$$

  - $U \sim$ Uniform$(-\pi/2, \pi/2)$
  - $W \sim$ Exp(1)
  - $\zeta = \dfrac{1}{\alpha} \arctan(\beta \tan(\pi\alpha/2))$

- Special care for $\alpha = 1$due to singularity in $\tan(\pi\alpha/2)$
- Efficient, standard method in libraries (e.g., levy_stable.rvs in SciPy)

LePage Series Representation

- Simulates $X = \sum_{j=1}^{\infty} \gamma_j \epsilon_j$
  - $\Gamma_j = E_1 + E_2 + \cdots + E_j$, with $E_j \sim$ Exp(1).
  - T_j are i.i.d Uniform$(0,1)$ or deterministic ($T_j = j$).
  - $\gamma_j = \left(\frac{\Gamma_j}{T_j}\right)^{-1/\alpha}$
  - For symmetric stable: $\epsilon_j \sim$ Uniform$(-1, +1)$
  - For general stable: $\epsilon_j = \frac{\sin(\alpha V_j)}{\cos(V_j)^{1/\alpha}}$, $V_j \sim$ Uniform$(-\pi/2, \pi/2)$
- Slow but theoretically insightful construction from Poisson point processes

FFT-Based Inversion of CF

- Choose a symmetric grid of t-values: $t_k = \dfrac{2\pi k}{T}$ for $k = -N/2, \ldots, N/2 - 1$.
- Compute $\phi(t_k)$from the known characteristic function of the stable law (Param A)
- Apply inverse FFT to obtain approximated PDF:

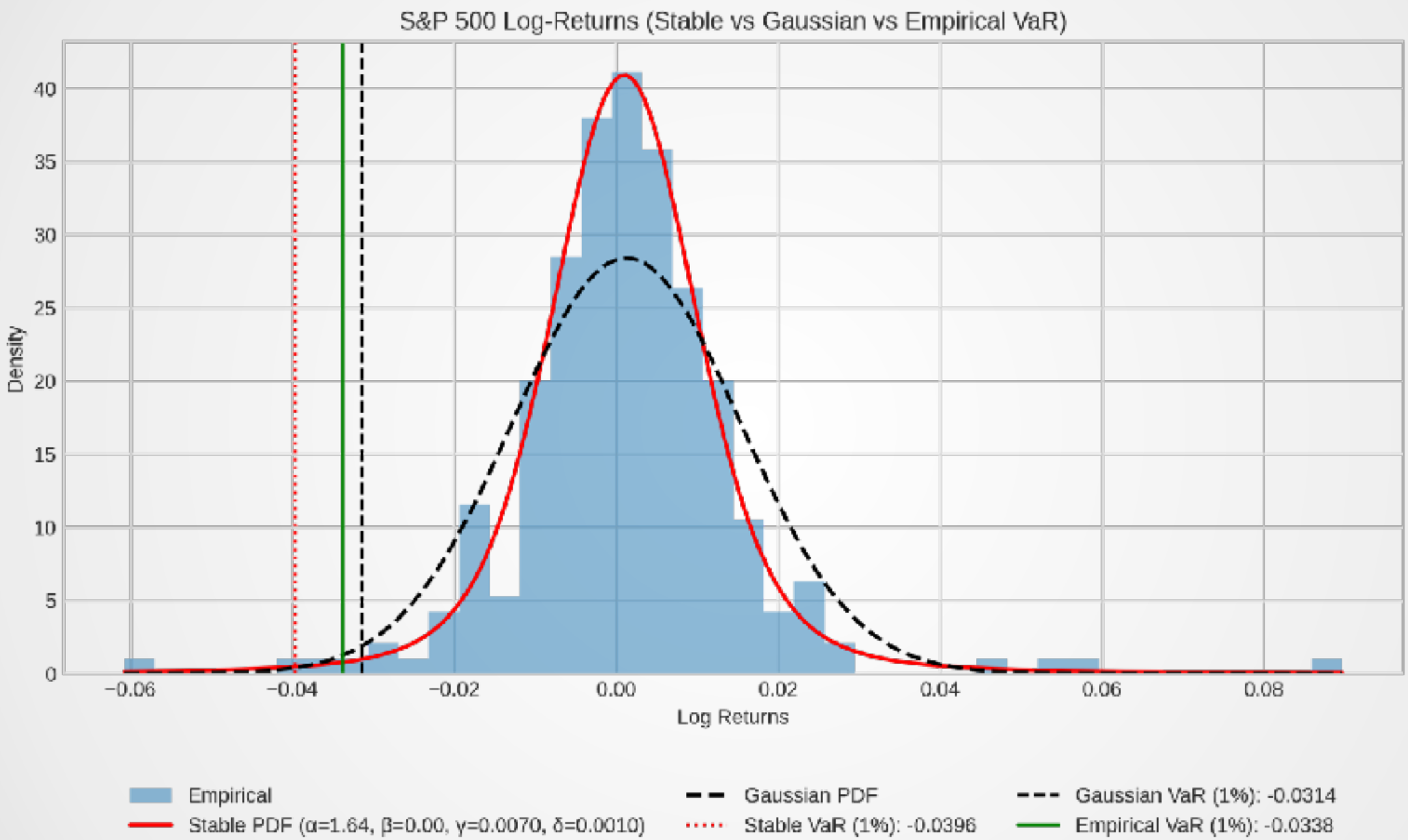
$$f(x) \approx \mathrm{Re}\left(\frac{1}{2\pi}\sum_k \phi\left(t_k\right) e^{-it_k x}\Delta t\right)$$

- Integrate PDF to obtain approximate CDF numerically
- Use inverse transform sampling: draw $U \sim$ Uniform$(0,1)$, invert $F(x)$.
- Suitable for generating large batches of samples for arbitrary $(\alpha, \beta, \gamma, \delta)$.
- High accuracy but computationally intensive (especially in high-resolution settings).

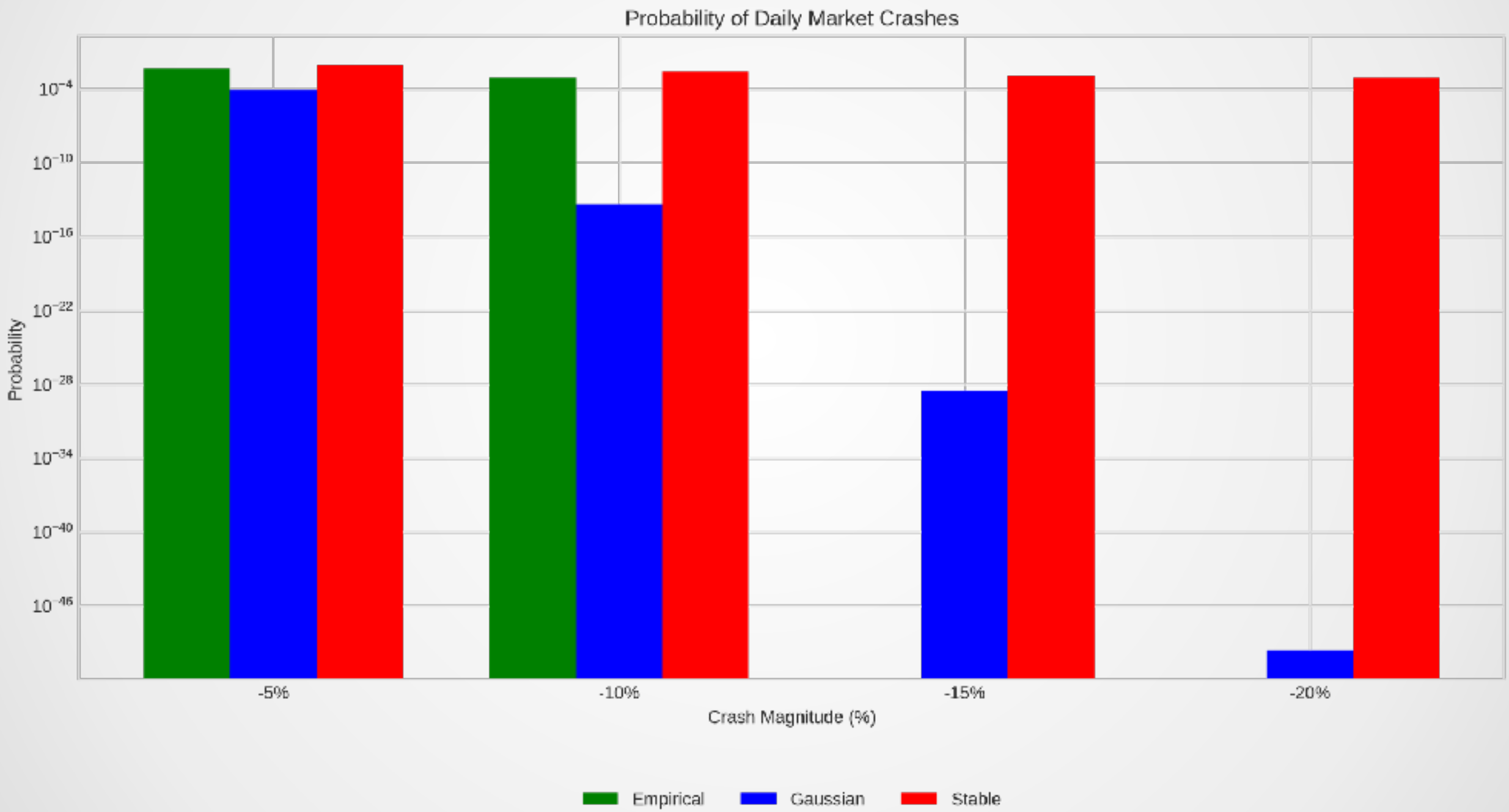
Practical Applications of Stable Distributions

- Finance and Risk Management
  - Model returns of financial assets with heavy tails and skewness
  - Replace Gaussian assumptions in portfolio risk models with α-stable alternatives
  - Better fit to empirical data for Value-at-Risk (VaR) and Expected Shortfall (ES) calculations
- Network Traffic and Telecommunications
  - Internet packet sizes and interarrival times exhibit heavy-tailed behavior
  - Stable models capture burstiness in traffic more accurately than Gaussian assumptions
  - Applied in modeling self-similar and long-range dependent network flows
- Signal and Image Processing
  - Impulsive noise models (e.g., radar, sonar, medical imaging)
- Physics and Environmental Sciences
  - Lévy flights in anomalous diffusion and statistical mechanics
  - Distribution of gravitational forces (Holtsmark) in astrophysics
  - Stable laws used to describe heavy rainfall, earthquake magnitudes

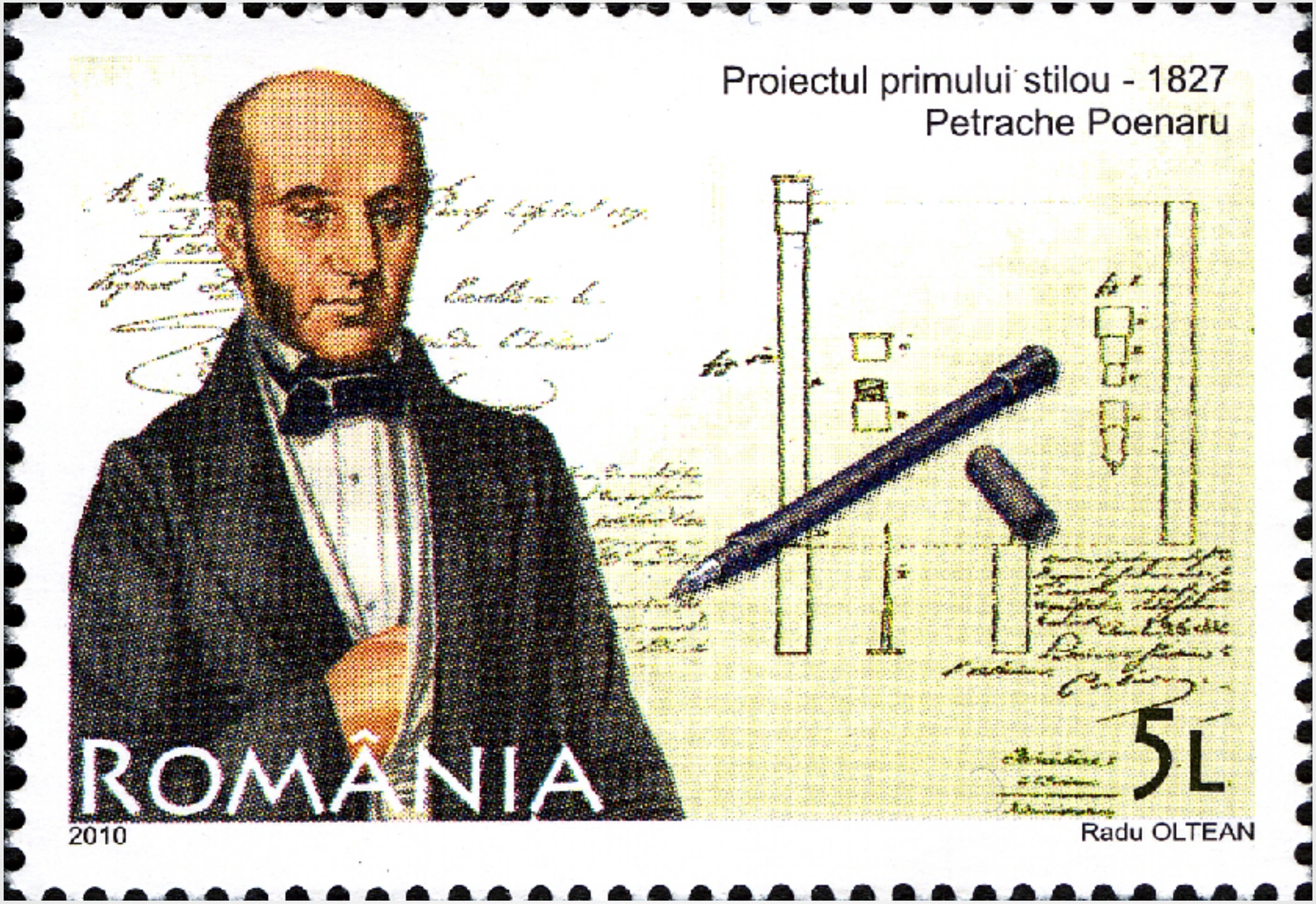
Slide Title


S&P 500 Log-Returns (Stable vs Gaussian vs Empirical VaR)
Density
Log Returns
0
5
10
15
20
25
30
35
40
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
Empirical
Stable PDF (α=1.64, β=0.00, γ=0.0070, δ=0.0010)
Gaussian PDF
Stable VaR (1%): -0.0396
Gaussian VaR (1%): -0.0314
Empirical VaR (1%): -0.0338

Slide Title


Probability of Daily Market Crashes
Probability
10^{-4}
10^{-10}
10^{-16}
10^{-22}
10^{-28}
10^{-34}
10^{-40}
10^{-46}
-5%
-10%
-15%
-20%
Crash Magnitude (%)
Empirical
Gaussian
Stable

Homeworks


Proiectul primului stilou - 1827
Petrache Poenaru
ROMÂNIA
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2010
Radu OLTEAN

Exercises and more

- **Problem 1: Theoretical Properties**

  a) Prove that if X_1 and X_2 are independent random variables from the same stable distribution $S(\alpha,0,1,0)$, then $X_1 + X_2$ follows $S(\alpha,0,2^{1/\alpha},0)$.

  b) Show that when $\alpha = 2$(normal distribution), the characteristic function reduces to that of a normal distribution.

  c) Explain why the variance of a stable distribution with $\alpha < 2$is infinite. Provide a mathematical argument.

- **Problem 2: Simulation and Visualization**

  a) Generate random samples from stable distributions with the following parameters:

  - $S(1.5,0,1,0)$
  - $S(1.8,0.5,1,0)$
  - $S(0.8,-0.3,1,0)$

  b) Generate 10,000 samples for each distribution and create histograms with overlaid theoretical PDFs.

  c) Plot the empirical tail distributions on a log-log scale and verify that they follow power laws with exponent $-\alpha$.

Exercises and more

- **Problem 3: Parameter Estimation**

  a) Generate 1,000 samples from a stable distribution with parameters $\alpha = 1.7$, $\beta = 0.2$, $\gamma = 1$, $\delta = 0$

  b) Implement the following parameter estimation methods:
  - Maximum likelihood estimation
  - Quantile-based estimation
  - Method based on characteristic function

  c) Compare the accuracy of each method and discuss which one performs best for your simulated data.

- **Problem 4: Financial Applications**

  a) Download at least 5 years of daily returns data for three different assets:
  - A stock market index (e.g., S&P 500)
  - An individual stock
  - A cryptocurrency

  b) For each dataset:
  - Fit a stable distribution
  - Fit a normal distribution
  - Compute the Kolmogorov-Smirnov test statistic for both distributions
  - Calculate VaR and Expected Shortfall at 1% and 5% levels for both distributions
  - Perform backtesting to evaluate which model better predicts tail events

  c) Write a brief report (1-2 pages) discussing your findings. Which asset shows the heaviest tails? How does this impact risk measures?

References

- Nolan, J. P. (2020). Stable Distributions: Models for Heavy-Tailed Data. Birkhäuser.
- Samorodnitsky, G., & Taqqu, M. S. (1994). Stable Non-Gaussian Random Processes. Chapman & Hall.
- Zolotarev, V. M. (1986). One-Dimensional Stable Distributions. AMS.
- Koutrouvelis, I. A. (1980). "Regression-type estimation of the parameters of stable laws." Journal of the American Statistical Association, 75(372), 918–928.
- McCulloch, J. H. (1986). "Simple consistent estimators of stable distribution parameters." Communications in Statistics - Simulation and Computation, 15(4), 1109–1136.
- Weron, R. (1996). "On the Chambers-Mallows-Stuck method for simulating skewed stable random variables." Statistics & Probability Letters, 28(2), 165–171.
- Feller, W. (1971). An Introduction to Probability Theory and Its Applications, Vol. 2. Wiley.